Prob. 3 (b)

$$\vec{E} = \frac{q}{4\pi\varepsilon_0}\left\{\frac{1-v^2/c^2}{\left(R-\frac{\vec{v}\cdot\vec{R}}{c}\right)^3}\left(\vec{R}-\frac{\vec{v}}{c}R\right) + \frac{\vec{R}\times\left(\vec{R}-\frac{\vec{v}}{c}R\right)\times\vec{a}}{c^2\left(R-\frac{\vec{v}\cdot\vec{R}}{c}\right)^3}\right\} \quad (1)$$

~~Now take the limit $\frac{v}{c} \ll 1$ and expand in that ratio~~

$$\vec{E} = \frac{q}{4\pi\varepsilon_0}\left\{\underbrace{\frac{(1-\beta^2)(\hat{R}-\vec{\beta})}{R^2(1-\vec{\beta}\cdot\hat{R})^3}}_{\vec{E}\text{ near-field}} + \underbrace{\frac{\hat{R}\times(\hat{R}-\vec{\beta})\times\vec{a}}{c^2R(1-\vec{\beta}\cdot\hat{R})^3}}_{\vec{E}\text{ far-field}}\right\} \quad (2)$$

Here $\hat{R} \equiv \frac{\vec{R}}{R}$, $\vec{\beta} = \frac{\vec{v}}{c}$, $\beta = \frac{v}{c}$.

Take the ratio $\frac{E_{near}}{E_{far}} \sim \frac{1}{R^2}\,\frac{c^2R}{a} = \frac{c^2}{Ra}$.

$\Rightarrow$ $E_{near} \sim E_{far}$ at $R_s \sim \frac{c^2}{a}$.

(i) For $R \ll R_s$ the field is dominated by the 1st term $\sim \frac{1}{R^2}$.

(ii) For $R \gg R_s$ the 2nd term dominates, $\sim \frac{a}{c^2R}$.

Eq. (2) can be expanded further by, for example, writing $\vec{\beta}\cdot\hat{R} = \beta\cos\theta$ (θ = angle between $\vec{v}$ and $\hat{R}$) and expanding in powers of β.

It is also worth noting that $\vec{E}_{far}\cdot\hat{R} = 0$, i.e. the far-field is transverse to $\hat{R}$, as of course it is should be for the field of electromagnetic wave.